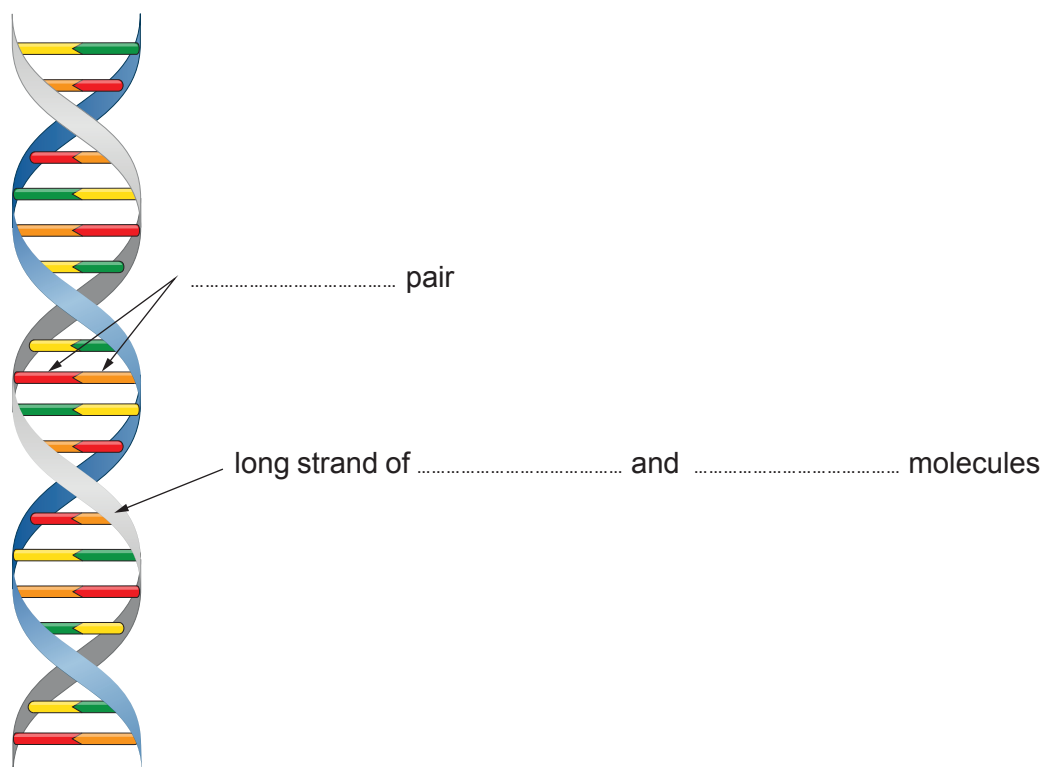


3. (a) **Image 3.1** shows part of a DNA molecule.

(i) **Complete the labels on Image 3.1** by choosing words from the box. [3]

phosphate	salt	base	sugar	amino acid
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Image 3.1



(ii) The shape of DNA is described as a twisted ladder. State the scientific term for this shape. [1]

.....

(b) The DNA of an individual can be analysed as a genetic profile.

Underline which **one** of the following correctly states how the DNA must first be treated in order to produce the bands in a genetic profile. [1]

Cut it into small pieces.

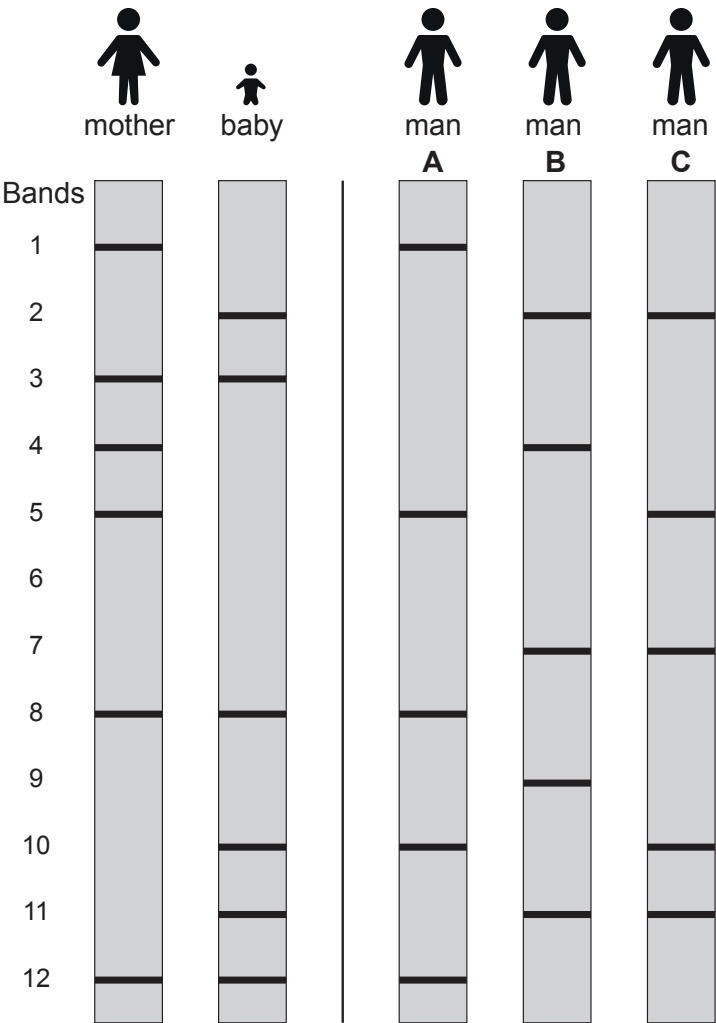
Place it in acid conditions.

Expose it to very low temperatures.



(c) **Image 3.2** shows the DNA profiles of people involved in a paternity test.

Image 3.2



- (i) In this baby's profile, three of the bands have come from the mother and three from the father.

Draw circles around the **three bands** in the **baby's profile** in **Image 3.2** which have come from the **mother**.

[1]

- (ii) Use the DNA profiles to conclude which man (**A**, **B** or **C**) is the baby's father. Explain how you obtained your answer.

[2]

.....

.....

.....

- (iii) Apart from paternity testing state **one** other use for DNA profiling.

[1]

.....

